

Stewart - Calculus 8e Chapter 13 Form C

Select the correct answer for each question.

- _____ 1. Find the limit.

$$\lim_{t \rightarrow 0^+} \langle 8 \cos t, 24 \sin t, 5t \ln t \rangle$$

- a. $\mathbf{r}(t) = 8\mathbf{i} + 24\mathbf{j} + 5\mathbf{k}$
- b. $\mathbf{r}(t) = 8\mathbf{k}$
- c. $\mathbf{r}(t) = 8\mathbf{i} - 5\mathbf{k}$
- d. $\mathbf{r}(t) = 8\mathbf{j}$
- e. $\mathbf{r}(t) = 8\mathbf{i}$

_____ 2. Let $\mathbf{r}(t) = \left\langle \sqrt{2-t}, \frac{(e^t - 2)}{t}, \ln(t+1) \right\rangle$.

Find the domain of r .

- a. $(-1, 0) \cup (0, 2]$
- b. $(2, \infty]$
- c. $[2, 0) \cup (0, 2)$
- d. $(-\infty, -1)$
- e. $(-1, 2]$

_____ 3. Find the domain of the vector function $\mathbf{r}(t) = \left\langle 8\sqrt{t}, \frac{1}{t-9}, \ln t \right\rangle$.

- a. $(0, 9) \cup (9, \infty)$
- b. $[0, 8) \cup (8, 9) \cup (9, \infty)$
- c. $(0, 8) \cup (8, \infty)$
- d. $(0, 8) \cup (8, 9) \cup (9, \infty)$

Stewart - Calculus 8e Chapter 13 Form C

4. Find a vector function that represents the curve of intersection of the two surfaces:

the top half of the ellipsoid $x^2 + 7y^2 + 7z^2 = 49$ and the parabolic cylinder $y = x^2$.

- a. $\mathbf{r}(t) = t\mathbf{i} + t^4\mathbf{j} + \sqrt{\frac{49 - t^2 + 7t}{7}}\mathbf{k}$
b. $\mathbf{r}(t) = t\mathbf{i} + t^2\mathbf{j} + \sqrt{\frac{49 + t^2 - 7t^4}{7}}\mathbf{k}$
c. $\mathbf{r}(t) = t\mathbf{i} + t^2\mathbf{j} + \sqrt{\frac{49 - t^2 - 7t}{7}}\mathbf{k}$
d. $\mathbf{r}(t) = t\mathbf{i} + t^2\mathbf{j} + \sqrt{\frac{7 - t^2 - 7t^4}{7}}\mathbf{k}$
e. $\mathbf{r}(t) = t\mathbf{i} - t^2\mathbf{j} - \sqrt{\frac{49 - t^2 - 7t}{7}}\mathbf{k}$

5. Find a vector function that represents the curve of intersection of the two surfaces:

The circular cylinder $x^2 + y^2 = 4$ and the parabolic cylinder $z = xy$.

- a. $\mathbf{r}(t) = 4\cos t\mathbf{i} + 4t\mathbf{j} + 4\cos^2 t\mathbf{k}$
b. $\mathbf{r}(t) = 2\cos(t)\mathbf{i} + 2\sin(t)\mathbf{j} - \sin(t)\cos(t)\mathbf{k}$
c. $\mathbf{r}(t) = \cos t\mathbf{i} + \sin t\mathbf{j} - 4\cos 2t\mathbf{k}$
d. $\mathbf{r}(t) = \cos(t)\mathbf{i} + \sin(t)\mathbf{j} + 4\sin(t)\cos(t)\mathbf{k}$
e. $\mathbf{r}(t) = 2\cos(t)\mathbf{i} + 2\sin(t)\mathbf{j} + 4\sin(t)\cos(t)\mathbf{k}$

6. If $\mathbf{r}(t) = \langle t, t^9, t^{11} \rangle$, find $\mathbf{r}''(t)$.

- a. $\langle 0, 72t^7, 110t^9 \rangle$
b. $\langle 0, 110t^9, 72t^7 \rangle$
c. $\langle 0, 72t^6, 110t^8 \rangle$
d. $\langle 1, 9t^6, 11t^8 \rangle$
e. $\langle 0, 110t^8, 72t^6 \rangle$

Stewart - Calculus 8e Chapter 13 Form C

- _____ 7. The helix $\mathbf{r}_1(t) = 4 \cos t \mathbf{i} + \sin t \mathbf{j} + t \mathbf{k}$ intersects the curve $\mathbf{r}_2(t) = (4+t)\mathbf{i} + 6t^2\mathbf{j} + 5t^3\mathbf{k}$ at the point $(4, 0, 0)$. Find the angle of intersection.
- a. 0
 - b. $\frac{\pi}{4}$
 - c. $\frac{\pi}{2}$
 - d. $\frac{\pi}{3}$
 - e. None of these
- _____ 8. Find parametric equations for the tangent line to the curve with parametric equations $x = 3t$, $y = 5t^2$, $z = 4t^3$ at the point with $t = 1$.
- a. $x = 1 + 3t$, $y = 1 + 10t$, $z = 1 + 12t$
 - b. $x = 3 + 3t$, $y = 5 + 10t$, $z = 4 + 12t$
 - c. $x = 1 + 3t$, $y = 1 + 5t$, $z = 1 + 4t$
 - d. $x = 3 + 3t$, $y = 5 + 5t$, $z = 4 + 4t$

Stewart - Calculus 8e Chapter 13 Form C

_____ 9. Find the unit tangent vector for the curve given by $\mathbf{r}(t) = \left\langle \frac{1}{5}t^5, \frac{1}{3}t^3, t \right\rangle$.

a. $\frac{\langle t^4, t^2, 1 \rangle}{\sqrt{t^{10} + t^4}}$

b. $\frac{\langle t^4, t^2, 1 \rangle}{\sqrt{4t^8 + 4t^4 + 1}}$

c. $\frac{\langle t^4, t^2, 1 \rangle}{\sqrt{t^8 + t^4 + 1}}$

d. $\frac{\langle t^4, t^2, 1 \rangle}{\sqrt{t^8 + t^4}}$

e. None of these

Stewart - Calculus 8e Chapter 13 Form C

- ____ 10. Reparametrize the curve with respect to arc length measured from the point where $t = 0$ in the direction of increasing t .

$$\mathbf{r}(t) = (7 + 3t)\mathbf{i} + (10 + 9t)\mathbf{j} - (6t)\mathbf{k}$$

a. $\mathbf{r}(t(s)) = \left(7 - \frac{3}{\sqrt{126}}s\right)\mathbf{i} + \left(10 + \frac{9}{\sqrt{126}}s\right)\mathbf{j} - \left(\frac{6s}{\sqrt{126}}\right)\mathbf{k}$

b. $\mathbf{r}(t(s)) = \left(7 + \frac{3}{\sqrt{126}}s\right)\mathbf{i} + \left(10 - \frac{9}{\sqrt{126}}s\right)\mathbf{j} - \left(\frac{6s}{\sqrt{126}}\right)\mathbf{k}$

c. $\mathbf{r}(t(s)) = \left(7 + \frac{3}{\sqrt{126}}s\right)\mathbf{i} + \left(10 + \frac{9}{\sqrt{126}}s\right)\mathbf{j} - \left(\frac{6s}{\sqrt{126}}\right)\mathbf{k}$

d. $\mathbf{r}(t(s)) = \left(7 + \frac{3}{\sqrt{126}}s\right)\mathbf{i} + \left(10 + \frac{9}{\sqrt{126}}s\right)\mathbf{j} + (6s)\mathbf{k}$

e. $\mathbf{r}(t(s)) = \left(7 + \frac{3}{\sqrt{126}}s\right)\mathbf{i} + \left(10 + \frac{9}{\sqrt{126}}s\right)\mathbf{j} - (6s)\mathbf{k}$

- ____ 11. Find the length of the curve $\mathbf{r}(t) = -3t\mathbf{i} + 2t\mathbf{j} - t\mathbf{k}$, $-2 \leq t \leq 1$.

a. $2\sqrt{14}$

b. $\sqrt{14}$

c. $6\sqrt{14}$

d. $3\sqrt{14}$

Stewart - Calculus 8e Chapter 13 Form C

- _____ 12. The torsion of a curve defined by $\mathbf{r}(t)$ is given by

$$\tau = \frac{(\mathbf{r}' \times \mathbf{r}'') \cdot \mathbf{r}'''}{|\mathbf{r}' \times \mathbf{r}''|^2}.$$

Find the torsion of the curve defined by $\mathbf{r}(t) = \cos 5t\mathbf{i} + \sin 5t\mathbf{j} + 4t\mathbf{k}$.

- a. $\frac{21}{41}$
- b. $\frac{9}{41}$
- c. $\frac{20}{41}$
- d. $\frac{80}{41}$
- _____ 13. Use Simpson's Rule with $n = 4$ to estimate the length of the arc of the curve with equations $x = \sqrt{t}$, $y = \frac{4}{t}$, $z = t^2 + 1$, from $(1, 4, 2)$ to $(2, 1, 17)$. Round your answer to four decimal places.
- a. 14.824
- b. 7.2041
- c. 15.8306
- d. 6.5706
- e. None of these

Stewart - Calculus 8e Chapter 13 Form C

- _____ 14. The curvature of the curve given by the vector function $\mathbf{r}$ is

$$k(t) = \frac{|\mathbf{r}'(t) \times \mathbf{r}''(t)|}{|\mathbf{r}'(t)|^3}$$

Use the formula to find the curvature of $\mathbf{r}(t) = \langle \sqrt{19}t, e^t, e^{-t} \rangle$ at the point $(0, 1, 1)$.

- a. $\sqrt{21}$
- b. $\frac{\sqrt{2}}{21}$
- c. $21\sqrt{2}$
- d. $\frac{21}{\sqrt{2}}$
- e. $\frac{\sqrt{21}}{21}$

- _____ 15. Find the velocity of a particle with the given position function.

$$\mathbf{r}(t) = 19e^{5t}\mathbf{i} + 5e^{-21t}\mathbf{j}$$

- a. $\mathbf{v}(t) = 19e^{5t}\mathbf{i} + e^{-21t}\mathbf{j}$
- b. $\mathbf{v}(t) = 95e^{5t}\mathbf{i} - 105e^{-21t}\mathbf{j}$
- c. $\mathbf{v}(t) = 19e^{5t}\mathbf{i} - 105e^{-21t}\mathbf{j}$
- d. $\mathbf{v}(t) = e^{5t}\mathbf{i} + 105e^{-21t}\mathbf{j}$
- e. $\mathbf{v}(t) = 95e^{5t}\mathbf{i} + 105e^{-21t}\mathbf{j}$

Stewart - Calculus 8e Chapter 13 Form C

- _____ 16. Find the speed of a particle with the given position function.

$$\mathbf{r}(t) = 4\sqrt{2}t\mathbf{i} + e^{4t}\mathbf{j} - e^{-4t}\mathbf{k}$$

a. $|\mathbf{v}(t)| = \sqrt{4 + 4e^t + 4e^{-t}}$

b. $|\mathbf{v}(t)| = (e^{4t} + e^{-4t})$

c. $|\mathbf{v}(t)| = \sqrt{4 + e^{4t} + e^{-40t}}$

d. $|\mathbf{v}(t)| = 4(e^{4t} + e^{-4t})$

e. $|\mathbf{v}(t)| = 4(e^t + e^{-t})$

- _____ 17. A mortar shell is fired with a muzzle speed of 250 ft/sec. Find the angle of elevation of the mortar if the shell strikes a target located 1100 ft away. Round your answer to 2 decimal places.

a. 17.14°

b. 14.69°

c. 0.3°

d. 0.9°

- _____ 18. Find the scalar tangential and normal components of acceleration of a particle with position vector $\mathbf{r}(t) = 4\sin t\mathbf{i} + 4\cos t\mathbf{j} + 3t\mathbf{k}$.

a. $a_{\mathbf{T}} = 0, a_{\mathbf{N}} = 4\sqrt{10}$

b. $a_{\mathbf{T}} = 3, a_{\mathbf{N}} = 4\sqrt{10}$

c. $a_{\mathbf{T}} = 3, a_{\mathbf{N}} = 4$

d. $a_{\mathbf{T}} = 0, a_{\mathbf{N}} = 4$

- _____ 19. A projectile is fired with an initial speed of 700 m/s and angle of elevation 60° . Find the range of the projectile.

a. $d \approx 43.3$ km

b. $d \approx 433$ km

c. $d \approx 350$ km

d. $d \approx 63.3$ km

e. $d \approx 53.3$ km

Stewart - Calculus 8e Chapter 13 Form C

- _____ 20. The following table gives coordinates of a particle moving through space along a smooth curve.

t	x	y	z
0.5	5.8	9.1	4.3
1	12.6	14.9	16.8
1.5	25.6	21.2	29.4
2	39.2	39.5	37.9
2.5	42.4	42.4	43

Find the average velocity over the time interval $[1, 2]$.

- a. $\mathbf{v} = 21.1\mathbf{i} + 24.6\mathbf{j} + 26.6\mathbf{k}$
- b. $\mathbf{v} = 21.1\mathbf{i} + 26.6\mathbf{j} + 21.1\mathbf{k}$
- c. $\mathbf{v} = 24.6\mathbf{i} + 24.6\mathbf{j} + 21.1\mathbf{k}$
- d. $\mathbf{v} = 26.6\mathbf{i} + 24.6\mathbf{j} + 21.1\mathbf{k}$
- e. $\mathbf{v} = 21.1\mathbf{i} + 26.6\mathbf{j} + 26.6\mathbf{k}$